CLASS	11^{th}	CHAPTER	7	EXERCISE	7.1

Q1: Prove the following propositions by mathematical induction for every positive integer n.

i. $1+3+5+\dots+(2n-1)=n^2$

Solution:

$1+3+5+\dots+(2n-1)=n^2$

i. For $n=1$, the given expression becomes

$$\Rightarrow (2(1)-1)=(1)^2$$
$$\Rightarrow (2-1)=1$$
$$\Rightarrow 1=1$$

$\therefore$ Since $P(n)$ is true for $n=1$

ii. Assume that the statement is true for $n=k$. i.e.,

$$= 1+3+5+\dots+(2k-1)=(k)^2$$
$$= 1+3+5+\dots+(2k-1)=k^2$$

iii. Now for $n=k+1$, we have,
Now, we have to prove the result is also true for $n=k+1$, To do so, we add $(2k+1)$ to the both sides of the equation:

$$1+3+5+\dots+(2k-1)+(2k+1)=k^2+2k+1$$
$$1+3+5+\dots+(2k-1)+(2k+1)=(k+1)^2$$
$$k^2+2k+1=(k+1)^2$$
$$(k+1)^2=(k+1)^2$$

$\therefore$ $P(n)$ is true for $n=k+1$.

Thus, by the princple of mathematical Induction $P(n)$ is true for all positive integers n.

ii. $3+6+9+\dots+3n=\frac{3}{2}n(n+1)$

Solution:

$3+6+9+\dots+3n=\frac{3}{2}n(n+1)$

i. For $n=1$, $P(n)$ becomes

$$\Rightarrow 3(1)=\frac{3}{2}(1)(1+1)$$
$$\Rightarrow 3=\frac{3}{2}(2)$$
$$\Rightarrow 3=3$$

So, $P(n)$ is true for $n=1$.

ii. Assuming the result to be true for $n=k$, we get the hypothesis:

$$\Rightarrow 3+6+9+\dots+3k=\frac{3}{2}k(k+1)$$

iii. Now we have to prove that the result is also true for $n=k+1$. To do so we add $3(k+1)$ to both sides of the above hypothesis, we get:

$$3+6+9+\dots+3k+3(k+1)=\frac{3}{2}k(k+1)+3(k+1)$$
$$\frac{3}{2}k(k+1)+3(k+1)=\frac{3}{2}k(k+1)+3(k+1)$$

$$\frac{3}{2}k(k+1) + 3(k+1) = \frac{3}{2}k(k+1) + 3(k+1)$$
$$3(k+1)\left\{\frac{k}{2} + 1\right\} = 3(k+1)\left\{\frac{k}{2} + 1\right\}$$
$$3(k+1)\left\{\frac{k+2}{2}\right\} = 3(k+1)\left\{\frac{k+2}{2}\right\}$$
$$\left\{\frac{3(k+1)((k+1)+1)}{2}\right\} = \left\{\frac{3(k+1)((k+1)+1)}{2}\right\}$$

i.e: $P(n)$ is true for $n = k+1$

Thus, by the principle of mathematical Induction $P(n)$ is true for all positive integers n.

iii. $1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$

Solution:

$$1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

i. For $n = 1$, $P(n)$ becomes.

$$= (3(1)-2) = \frac{1(3(1)-1)}{2}$$
$$= 3-2 = \frac{3-1}{2}$$
$$= 1 = \frac{\cancel{2}}{\cancel{2}}$$
$$= 1 = 1$$

so, $P(n)$ is true for $n = 1$.

ii. Assuming the result to be true for $n = k$, we get the hypothesis:

$$1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2}$$

iii. Now, we have to prove that the result is also true for $n = k+1$. To do so add $(3k+1)$ on both sides of the above hypothesis:

$$1 + 4 + 7 + \dots + (3k-2) + (3k+1) = \frac{k(3k-1)}{2} + (3k+1)$$
$$\frac{k(3k-1)}{2} + (3k+1) = \frac{k(3k-1)}{2} + (3k+1)$$
$$\frac{k(3k-1) + 2(3k+1)}{2} = \frac{k(3k-1) + 2(3k+1)}{2}$$
$$\frac{3k^2 - k + 6k + 2}{2} = \frac{3k^2 - k + 6k + 2}{2}$$
$$\frac{3k^2 + 5k + 2}{2} = \frac{3k^2 + 5k + 2}{2}$$
$$\frac{3k^2 + 3k + 2k + 2}{2} = \frac{3k^2 + 3k + 2k + 2}{2}$$
$$\frac{3k(k+1) + 2(k+1)}{2} = \frac{3k(k+1) + 2(k+1)}{2}$$
$$\frac{(3k+2)(k+1)}{2} = \frac{(3k+2)(k+1)}{2}$$

$$\frac{3k^2-k+6k+2}{2} = \frac{3k^2-k+6k+2}{2}$$

$$\frac{3k^2+5k+2}{2} = \frac{3k^2+5k+2}{2}$$

$$\frac{3k^2+3k+2k+2}{2} = \frac{3k^2+3k+2k+2}{2}$$

$$\frac{3k(k+1)+2(k+1)}{2} = \frac{3k(k+1)+2(k+1)}{2}$$

$$\frac{(3k+2)(k+1)}{2} = \frac{(3k+2)(k+1)}{2}$$

$$\frac{(k+1)(3k+3-1)}{2} = \frac{(k+1)(3k+3-1)}{2}$$

$$\frac{(k+1)(3(k+1)-1)}{2} = \frac{(k+1)(3(k+1)-1)}{2}$$

$\therefore$ $P(n)$ is true for $n = k+1$

Thus, By the principle of mathematical Induction $P(n)$ is true for all positive integers.

iv. $1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} = 2\left(1 - \frac{1}{2^n}\right)$

Solution:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} = 2\left(1 - \frac{1}{2^n}\right)$$

i. For $n = 1$, $P(n)$ becomes

$$\frac{1}{2^{1-1}} = 2\left(1 - \frac{1}{2^1}\right)$$

$$\frac{1}{2^0} = \cancel{2}\left(\frac{2-1}{\cancel{2}}\right) \qquad 2^0 = 1$$

$1 = 1$ So, $P(n)$ is true for $n = 1$.

ii. Assuming the result to be true for $n = k$, we get the hypothesis:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{k-1}} = 2\left(1 - \frac{1}{2^k}\right)$$

iii. Now, we have to prove that the result is also true for $n = k+1$. To do so add $\frac{1}{2^k}$ on both sides of the above hypothesis:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{k-1}} + \frac{1}{2^k} = 2\left(1 - \frac{1}{2^k}\right) + \frac{1}{2^k}$$

$$= 2\left(\frac{2^k - 1}{2^k}\right) + \frac{1}{2^k}$$

$$= 2 - \frac{2}{2^k} + \frac{1}{2^k}$$

$$= 2 - \frac{1}{2^k}$$

$$= 2\left(1 - \frac{1}{2.2^k}\right)$$

$\therefore$ $P(n)$ is true for $n = k+1$ $= 2\left(1 - \frac{1}{2^{k+1}}\right)$

Thus, By the principle of mathematical Induction $P(n)$ is true for all positive integers.

v. $2+6+18+\ldots+2.3^{n-1}=3^n-1$

Solution:

$2+6+18+\ldots+2\cdot3^{n-1}=3^n-1$

i. For $n=1$, $P(n)$ becomes

$$2.3^{1-1}=3^1-1 \qquad \because 3^0=1$$
$$2.3^0=3-1$$
$$2=2$$

So, $P(n)$ is true for $n=1$

ii. Assuming the result to be true for $n=k$, we get following hypothesis:

$$2+6+18+\ldots+2.3^{k-1}=3^k-1$$

iii. Now, we have to prove that the result is also true for $n=k+1$, To do so add 2.3^k to both sides, of the above hypothesis and get.

$$2+6+18+\ldots+2.3^{k-1}+2.3^k=3^k-1+2.3^k$$
$$=3^k+2.3^k-1$$
$$=3^k(1+2)-1$$
$$=3.3^k-1$$
$$=3^{k+1}-1$$

So, it is true true for $n=k+1$

Thus, by the principle of mathematical Induction $P(n)$ is true for all positive integers n.

Mathematical Induction: Mathematical Statement ko proof karny K liye

First n natural number:-

$\Sigma 50 = n(n+1)$

Deduction: General case → Specific case.

Induction: Specific case → General case.

The priciple of Mathematical Induction:-

for a statement $P(n)$ involving the natural number n, such that

... the truth of $P(k+1)$

vi- $2+6+12+\dots+n(n+1)=\frac{1}{3}n(n+1)(n+2)$

Solution:

$$2+6+12+\dots+n(n+1)=\frac{1}{3}n(n+1)(n+2)$$

i. If $P(n)$ respresent the above proposition then.

For $n=1$ then $P(n)$ becomes.

$$1(1+1)=\frac{1}{3}(1)(1+1)(1+2)$$

$$2=\frac{1}{\cancel{3}}(2)(\cancel{3})$$

$$2=2$$ So, $P(n)$ is true for $n=1$

ii. Assuming the result to be true for $n=k$, we get the hypothesis:

$$2+6+12+\dots+k(k+1)=\frac{1}{3}k(k+1)(k+2)$$

iii. We have to prove that the result is also true for $n=k+1$, To do so weadd $(k+1)(k+2)$ on both sides of the above hypothesis:

k+1 term = k(k+1)
(k+1)(k+1+1)
(k+1)(k+2)

$$2+6+12+\dots+k(k+1)+(k+1)(k+2)=$$

$$\frac{1}{3}k(k+1)(k+2)+(k+1)(k+2)$$

$$=\frac{\frac{1}{3}k(k+1)(k+2)+3(k+1)(k+2)}{3}$$

$$=\frac{1}{3}(k+1)(k+2)\{k+3\}$$

$$=\frac{1}{3}(k+1)(k+2)(k+3)$$

$$=\frac{1}{3}(k+1)(k+1+1)(k+1+2)$$

I.e: $P(n)$ is true for $n=k+1$.

Thus by the principle of mathematical Induction $P(n)$ is true for all positive integers.

viii. $2^2 + 4^2 + 6^2 + \dots + (2n)^2 = \frac{2}{3}n(n+1)(2n+1)$

Solution:

$2^2 + 4^2 + 6^2 + \dots + (2n)^2 = \frac{2}{3}n(n+1(2n+1)$

i. If $P(n)$ represent the above proposition then,

Put $n=1$, $P(n)$ becomes:

k+1 term $= (2k)^2$

$[2(k+1)]^2$

$4(k+1)^2$

$$(2(1))^2 = \frac{2}{3}(1)(1+1)(2\times1+1)$$

$$(2)^2 = \frac{2}{\cancel{3}}(2)(\cancel{3})$$

$$4 = 4$$ So, $P(n)$ is true for $n=1$.

ii. Assuming the result to be true for $n=K$ we get the hypothesis:

$$2^2 + 4^2 + 6^2 + \dots + (2K)^2 = \frac{2}{3}K(K+1)(2K+1)$$

iii. Now we have to prove that the result is also true for $n=k+1$, To do so, We have add $4(K+1)^2$ on both sides on the above hypothesis:

$$2^2 + 4^2 + 6^2 + \dots (2K)^2 + 4(K+1)^2 = \frac{2}{3}K(K+1)(2K+1) + 4(K+1)^2$$

$$= \frac{2}{3}K(K+1)(2K+1) + 4(K+1)^2$$

$$= 2(K+1)\left\{\frac{1}{3}K(2K+1) + 2(K+1)\right\}$$

$$= 2(K+1)\left\{\frac{K(2K+1) + 6(K+1)}{3}\right\}$$

$$= \frac{2}{3}(K+1)\left\{K(2K+1) + 6(K+1)\right\}$$

$$= \frac{2}{3}(K+1)\left\{2K^2 + K + 6K + 6\right\}$$

$$= \frac{2}{3}(K+1)\left\{2K^2 + 7K + 6\right\}$$

$$= \frac{2}{3}(K+1)\left\{2K^2 + 4K + 3K + 6\right\}$$

$$= \frac{2}{3}(K+1)\left[2K(K+2) + 3(K+2)\right\}$$

$$= \frac{2}{3}(K+1)(K+2)(2K+3)$$

$$= \frac{2}{3}(K+1)(K+1+1)(2K+2+1)$$

$$= \frac{2}{3}\left[(K+1)(K+1+1)(2(K+1)+1)\right]$$

$\therefore P(n)$ is true for $n=k+1$

Thus, by the principle of mathematical Induction $P(n)$ is true for all positive integers n.

viii. $1.3 + 2.4 + 3.5 + \dots + n(n+2) = \frac{1}{6}n(n+1)(2n+7)$.

Solution: If $P(n)$ represent the above proposition then ~~Suppose that~~ For $n=1$. $P(n)$ becomes.

$$1(1+2) = \frac{1}{6}(1)(1+1)(2(1)+7)$$

$$3 = \frac{1}{\cancel{6}}(2)(9)^{3}$$

$$3 = 3$$

So, $P(n)$ is true for $n=1$.

ii. Assuming the result to be true for $n=k$, we get the hypothesis.

$$1.3 + 2.4 + 3.5 + \dots + k(k+2) = \frac{1}{6}k(k+1)(2k+7)$$

iii. Now, we have to prove the result is true for $n=k+1$, To do so, we add $(k+1)(k+3)$ to both sides of the above hypothesis, we get.

$$1.3 + 2.4 + 3.5 + \dots + k(k+2) + (k+1)(k+3) = \frac{1}{6}k(k+1)(2k+7) + (k+1)(k+3)$$

$$= \frac{1}{6}k(k+1)(2k+7) + (k+1)(k+3)$$

$$= \frac{k(k+1)(2k+7) + 6(k+1)(k+3)}{6}$$

$$= \frac{(k+1)}{6}\{k(2k+7) + 6(k+3)\}$$

$$= \frac{k+1}{6}\{2k^2 + 7k + 6k + 18\}$$

$$= \frac{k+1}{6}\{2k^2 + 13k + 18\}$$

$$= \frac{k+1}{6}[(2k^2 + 4k + 9k + 18)]$$

$$= \frac{k+1}{6}[2k(k+2) + 9(k+2)]$$

$$= \frac{k+1}{6}(k+2)(2k+9)$$

$$= \frac{k+1}{6}(k+1+1)(2k+2+7)$$

$$= \frac{k+1}{6}(k+1+1)(2(k+1)+7)$$

$P(n)$ is true for $n=k+1$.

Thus, by the principle of mathematical Induction $P(n)$ is true for all positive integers n.

ix. $\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$

Solution:

$$\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$$

i. If $P(n)$ represents the above proportion then

Put $n=1$, $P(n)$ becomes,

$$\frac{1}{1(1+1)}=\frac{1}{1+1}$$

$$\frac{1}{2}=\frac{1}{2}$$

So, $P(n)$ is true for $n=1$

ii. Assuming the result to be true for $n=k$, we get the hypothesis:

$$\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\cdots+\frac{1}{k(k+1)}=\frac{k}{k+1}$$

iii. Now, we have to Prove that the result is also true for $n=k+1$. To do so, we add $\frac{1}{(k+1)(k+2)}$ on both sides of the above hypothesis:

$$\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\cdots+\frac{1}{(k)(k+1)}+\frac{1}{(k+1)(k+2)}=\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k(k+2)+1}{(k+1)(k+2)}$$

$$=\frac{k^2+2k+1}{(k+1)(k+2)}$$

$$=\frac{k^2+k+k+1}{(k+1)(k+2)}$$

$$=\frac{k(k+1)+1(k+1)}{(k+1)(k+2)}$$

$$=\frac{\cancel{(k+1)}(k+1)}{\cancel{(k+1)}(k+2)}$$

$$=\frac{k+1}{(k+1+1)}$$

It is true for $n=k+1$

Hence, by the principle of mathematical Induction $P(n)$ is true for ~~not~~ all positive integers n.

x- $1.2 + 2.2^2 + 3.2^2 + 4.2^2 + \ldots + n.2^n = (n-1).2^{n+1} + 2$

Solution:

If $P(n)$ represents the above proposition then:-

For $n=1$, $P(n)$ becomes,

$$1.2^1 = (1-1).2^{1+1} + 2$$

$$2 = 0.2^2 + 2$$

$$2 = 2$$ So, $P(n)$ is true for $n=1$

Assuming the result to be true for $n=k$, we get the hypothesis:

$$1.2 + 2.2^2 + 3.2^2 + \ldots + k.2^k = (k-1).2^{k+1} + 2$$

Now, we have to prove that the result is true for $n = k+1$. To do so we add $(k+1).2^{k+1}$ to both sides of the above hypothesis:

kth term $k.2^k$. $(k+1).2^{k+1}$

$$1.2 + 2.2^2 + 3.2^2 + \ldots + k.2^k + (k+1).2^{k+1} =$$

$$(k-1).2^{k+1} + 2 + (k+1).2^{k+1}$$

$$= (k-1).2^{k+1} + (k+1).2^{k+1} + 2$$

$$= 2^{k+1}[(k-1) + (k+1)] + 2$$

$$= 2^{k+1}(k-\cancel{1} + k+\cancel{1}) + 2$$

$$= 2^{k+1}(2k) + 2$$

$$= 2.2^{k+1}(k) + 2$$

$$= 2^{k+1+1}(k+1-1) + 2$$

$$= (k+1-1).2^{k+1+1} + 2$$

$P(n)$ is true for $n = k+1$.

Thus, By the Principle of mathematical Induction $P(n)$ is true for all positive integers n.

xi. $1!.1 + 2!.2 + 3!.3 + \dots + n!.n = (n+1)! - 1$.

Solution:

If given proposition represent $P(n)$ then for $n = 1$, $P(n)$ becomes,

$$1!.1 = (1+1)! - 1$$
$$1 = 2! - 1$$
$$1 = 2 - 1$$
$$1 = 1$$

so $P(n)$ is true for $n = 1$

Assuming the result to be true for $n = k$, we get the hypothesis:

$$1!.1 + 2!.2 + 3!.3 + \dots + k!.k = (k+1)! - 1$$

Now, we have to prove that the result is also true for $n = k+1$, To do so, we add $(k+1)!.(k+1)$ to both sides of the above hypothesis then, we get,

$$1!.1 + 2!.2 + 3!.3 + \dots + k!.k + (k+1)!.(k+1) = (k+1)! - 1 + (k+1)!(k+1)$$
$$= (k+1)! - 1 + (k+1)!.(k+1)$$
$$= (k+1)! + (k+1)!.(k+1) - 1$$
$$= (k+1)!.(1 + k + 1) - 1$$
$$= (k+1)!.(k+2) - 1$$
$$= (k+2)(k+1)! - 1$$
$$= (k+2)! - 1$$
$$= (k+1+1)! - 1$$

$\therefore P(n)$ is true for $n = k+1$.

Hence, by the principle of mathematical Induction $P(n)$ is true for all positive integers n.

xiii - $\dfrac{1}{a(a+1)} + \dfrac{1}{(a+1)(a+2)} + \cdots + \dfrac{1}{(a+n-1)(a+n)} = \dfrac{n}{a(a+n)}$

Solution:

$$\frac{1}{a(a+1)} + \frac{1}{(a+1)(a+2)} + \cdots + \frac{1}{(a+n-1)(a+n)} = \frac{n}{a(a+n)}$$

If given proposition represent $P(n)$ then,

For $n=1$, $P(n)$ becomes,

$$\frac{1}{(a+\cancel{1}-\cancel{1})(a+1)} = \frac{1}{a(a+1)}$$

$$\frac{1}{a(a+1)} = \frac{1}{a(a+1)}$$

So, $P(n)$ is true for $n=1$.

Assuming the result to be true for $n=k$, we get the hypothesis:

$$\frac{1}{a(a+1)} + \frac{1}{(a+1)(a+2)} + \cdots + \frac{1}{(a+k-1)(a+k)} = \frac{k}{a(a+k)}.$$

Now, we have to prove that the result is also true for $n=k+1$. To do so we add $\dfrac{1}{(a+k)(a+k+1)}$ to both sides of the above hypothesis

$$\frac{1}{a(a+1)} + \frac{1}{(a+1)(a+2)} + \cdots + \frac{1}{(a+k-1)(a+k)} + \frac{1}{(a+k)(a+k+1)} = \frac{k}{a(a+k)} + \frac{1}{(a+k)(a+k+1)}$$

$$= \frac{k}{a(a+k)} + \frac{1}{(a+k)(a+k+1)}$$

$$= \frac{k(a+k+1) + a}{a(a+k)(a+k+1)}$$

$$= \frac{ka + k^2 + k + a}{a(a+k)(a+k+1)}$$

$$= \frac{k(a+k) + 1(a+k)}{a(a+k)(a+k+1)}$$

$$= \frac{\cancel{(a+k)}(k+1)}{a\cancel{(a+k)}(a+k+1)}$$

$$= \frac{(k+1)}{a(a+k+1)}$$

$P(n)$ is true for $n = k+1$

Proved

xiii- $\dfrac{1^2}{1.3}+\dfrac{2^2}{3.5}+\dots+\dfrac{n^2}{(2n-1)(2n+1)}=\dfrac{n(n+1)}{2(2n+1)}$.

Solution:

$$\frac{1^2}{1.3}+\frac{2^2}{3.5}+\dots+\frac{n^2}{(2n-1)(2n+1)}=\frac{n(n+1)}{2(2n+1)}$$

If $P(n)$ represent the above proposition then.
For $n=1$, $P(n)$ becomes,

k th term =

$$\frac{k^2}{(2k-1)(2k+1)} = \frac{(k+1)^2}{(2(k+1)-1)(2(k+1)+1)}$$

$$\frac{(k+1)^2}{(2k+2-1)(2k+2+1)}$$

$$\frac{(k+1)^2}{(2k+1)(2k+3)}$$

$$\frac{1^2}{(2(1)-1)(2(1)+1)}=\frac{1(1+1)}{2(2(1)+1)}$$

$$\frac{1}{(2-1)(2+1)}=\frac{\cancel{2}}{\cancel{2}(3)}$$

$$\frac{1}{3}=\frac{1}{3}$$

, $P(n)$ is true for $n=1$.

Assuming the result to be true for $n=k$. We get the hypothesis:

$$=\frac{1^2}{1.3}+\frac{2^2}{3.5}+\dots+\frac{k^2}{(2k-1)(2k+1)}=\frac{k(k+1)}{2(2k+1)}$$

- Now, we have to prove that the result is also true for $n=k+1$, To do so, we add $\dfrac{(k+1)^2}{(2k+1)(2k+3)}$ to both Sides. in the above hypothesis:

$$\frac{1^2}{1.3}+\frac{2^2}{3.5}+\dots+\frac{k^2}{(2k-1)(2k+1)}+\frac{(k+1)^2}{(2k+1)(2k+3)}=\frac{k(k+1)}{2(2k+1)}+\frac{(k+1)^2}{(2k+1)(2k+3)}$$

$$=\frac{k(k+1)(2k+3)+2(k+1)^2}{2(2k+1)(2k+3)}$$

$$=\frac{(k^2+k)(2k+3)+2(k^2+2k+1)}{2(2k+1)(2k+3)}$$

$$=\frac{(k+1)\left[k(2k+3)+2(k+1)\right]}{2(2k+1)(2k+3)}$$

$$=\frac{(k+1)(2k^2+3k+2k+2)}{2(2k+1)(2k+3)}$$

$$=\frac{(k+1)(2k^2+5k+2)}{2(2k+1)(2k+3)}$$

$$= \frac{(k+1)(2k^2+4k+k+2)}{2(2k+1)(2k+3)}$$

$$= \frac{(k+1)[2k(k+2)+1(k+2)]}{2(2k+1)(2k+3)}$$

$$= \frac{(k+1)(k+2)\cancel{(2k+1)}}{2\cancel{(2k+1)}(2k+3)}$$

$$= \frac{(k+1)(k+1+1)}{2(2k+2+1)}$$

$$= \frac{(k+1)(k+1+1)}{2[2(k+1)+1]}$$

$P(n)$ is true for $n = k+1$.

Hence, the principle of mathematical Induction is true for all positive integers n.

xiv-(i) $a + ar + ar^2 + \dots + ar^{n-1} = \dfrac{a(1-r^n)}{1-r}, (r \neq 1)$

Solution:

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

If $P(n)$ represent the above Proportion then, For $n=1$, $P(n)$ becomes,

$$ar^{1-1} = \frac{a(1-r^1)}{1-r}$$

$$ar^0 = \frac{a\cancel{(1-r)}}{\cancel{1-r}} \qquad \because r^0 = 1$$

$$ar^0 = a$$

$a = a$ So, $P(n)$ is true for $n=1$

Assuming the result to be true for $n=k$, we get the hypothesis:

$$a + ar + ar^2 + \dots + ar^{k-1} = \frac{a(1-r^k)}{1-r}$$

Now, we have to prove that the result is also true for $n=k+1$. To do so we add ar^k to both sides in the above hypothesis, we get.

$$a + ar + ar^2 + \dots + ar^{k-1} + ar^k = \frac{a(1-r^k)}{1-r} + ar^k.$$

$$= \frac{a(1-r^k) + ar^k(1-r)}{(1-r)}$$

$$= \frac{a(1-\cancel{r^k}+\cancel{r^k}-r.r^k}{(1-r)}$$

$$= \frac{a(1-r^{k+1})}{(1-r)}$$

$\therefore$ $P(n)$ is true for $n=k+1$

Thus, By the principle of mathematical Induction $P(n)$ is true for all positive integer n.

xiv- (iii)- $\binom{n}{0}+\binom{n}{1}+\dots+\binom{n}{n}=2^n$.

Solution:

$\binom{n}{0}+\binom{n}{1}+\dots+\binom{n}{n}=2^n$

If $P(n)$ represent the above proposition then,
for $n=1$, we get $P(n)$.

$\binom{1}{0}+\binom{1}{1}=2^1$

$\binom{n}{r}=\frac{n!}{r!(n-r)!}$

$\frac{1!}{0!(1-0)!}+\frac{1!}{1!(1-1)!}=2$

$\frac{\cancel{1!}}{1\,\cancel{1!}}+\frac{\cancel{1!}}{\cancel{1!}\,1}=2$

$1+1=2$

$2=2$

So, $P(n)$ is true for $n=1$.

Assuming that the result is true for $n=k$, we get the hypothesis.

$\binom{k}{0}+\binom{k}{1}+\dots+\binom{k}{k}=2^k$

Now, we have to prove that the ~~tr~~ result is true for above hypothesis:

$n=k+1$, So put $n=k+1$ in above hypothesis:

$=\binom{k+1}{0}+\binom{k+1}{1}+\dots\binom{k+1}{k+1}$ $\quad\because \binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$

$=\binom{k+1}{0}+\left[\binom{k+1-1}{1}+\binom{k+1-1}{1-1}\right]+\left[\binom{k+1-1}{2}+\binom{k+1-1}{2-1}\right]+\dots$

$\dots+\left[\binom{k+1}{k+1}\right]$ $\quad \binom{n+1}{n+1}=\binom{n}{n}=1 \quad \because \binom{n+1}{0}=1=\binom{n}{0}$

$=\binom{k+1}{0}+\binom{k}{1}+\binom{k}{0}+\binom{k}{2}+\binom{k}{1}+\dots+\binom{k+1}{k+1}$

$\binom{k}{k}$

$=\binom{k}{0}+\binom{k}{1}+\binom{k}{0}+\binom{k}{2}+\binom{k}{1}+\dots\binom{k}{k}$

$=2\binom{k}{0}+2\binom{k}{1}+2\binom{k}{2}+\dots2\binom{k}{k}$

$$= 2\left[\binom{k}{0} + \binom{k}{1} + \binom{k}{2} + \cdots + \binom{k}{k}\right]$$

$$= 2\left(2^{k}\right)$$

$$= 2^{k+1}$$

$P(n)$ is true for $n = k+1$

Thus, By the Principle of mathematical Induction $P(n)$ is true for every positive integer n.

xiv (iii). $\binom{n}{r-1}+\binom{n}{r}=\binom{n+1}{r}$; $n \geq r$

Solution:

$$\binom{n}{r-1}+\binom{n}{r}=\binom{n+1}{r}$$

If $P(n)$ represent the above proposition then,

$$\binom{n}{r-1}+\binom{n}{r}=\binom{n+1}{r}$$

For $n=1$, $P(n)$ becomes,

$$\binom{1}{r-1}+\binom{1}{r}=\binom{1+1}{r}$$

$$\binom{1+1}{r}=\binom{1+1}{r}$$

$$\binom{2}{r}=\binom{2}{r}$$ So, $P(n)$ is true for $n=1$.

Assuming that the result is true for $n=k$, we get hypothesis:

$$\binom{k}{r-1}+\binom{k}{r}=\binom{k+1}{r}$$

Now, we have to prove that the result is true for $n=k+1$

$$\binom{k+1}{r-1}+\binom{k+1}{r}=\binom{k+1+1}{r}$$

$$=\binom{k+2}{r}$$

$P(n)$ is true for $n=k+1$

Thus, the principle of mathematical Induction is true for every positive integer n.

Q#2: Prove the following statement by mathematical Induction:

i- $2^{3n+2} - 28n - 4$ is divisible by 49, $\forall\, n \in N$

Solution:

If $P(n)$ represent the above proposition, then for $n=1$, $P(n)$ becomes.

$= 2^{3\times1+2} - 28(1) - 4$

$= 2^5 - 28 - 4$

$= 32 - 32$

$= 0$

$2\times2\times2\times2\times2 = 32$

Since, 0 is divisible by 49, the given statement is true for $n=1$.

Assuming the statement is true for $n=k$ we get the hypothesis.

$2^{3k+2} - 28k - 4$ is divisible by 49

By our hypothesis $(2^{3k+2} - 28k - 4)$ is divisible by 49 then, Put $n=k+1$, we have,

Now, We have to prove that the result is true for $n=k+1$.

$2^{3(k+1)+2} - 28(k+1) - 4$

$2^{3k+3+2} - 28k - 28 - 4$

$2^3 . 2^{3k+2} - 28k - 32$

$8 . 2^{3k+2} - 28k - 32 + 196k - 196k$

$8 . 2^{3k+2} - 224k - 32 + 196k$

$8 . 2^{3k+2} - 28k - 4\} + 196k$

$8\{2^{3k+2} - 28k - 4\}$

By our hypothesis $(2^{3k+2} - 28k - 4)$ is divisible by 49 and $196k$ is obviously divisible by 49. Therefore the given expression is also divisible by 49 for $n=k+1$.

Hence, by mathematical Induction the given statement is true for all natural number.

ii. $3^{2n+2} - 8n - 9$ is divisible by 64, $\forall n \in N$

Solution:

If $P(n)$ represent above proposition then for $n=1$, $P(n)$ becomes,

$= 3^{2\times1+2} - 8(1) - 9$

$3\times3\times3\times3$

$= 3^{2+2} - 8 - 9$

$= 3^4 - 8 - 9$

$= 81 - 17$

$= 64$ Since 64 is divisible by 64, the given statement is true for $n=1$.

Assuming that the result is true for $n=k$, we get the hypothesis:

$3^{2k+2} - 8k - 9$ is exactly divisible by 64.

Now, we have to prove that the result is also true for $n = k+1$.

$3^{2(k+1)+2} - 8(k+1) - 9$

$3^{2k+2+2} - 8k - 8 - 9$

$3^2 . 3^{2k+2} - 8k - 17$

$9.3^{2k+2} - 8k - 17 + 64k - 64k + 64 - 64$

$9.3^{2k+2} - 72k - 81 + 64k + 64$

$9\{3^{2k+2} - 8k - 9\} + 64k + 64$

By our hypothesis $(3^{2k+2} - 8k - 9)$ is divisible by 64 and $64k + 64$ is obviously divisible by 64. Therefore the expression is also divisible by 64 for $n = k+1$.

Hence, By mathemical Induction the given statement is true for all Natural numbers.

iii. $7^n - 4^n$ is divisible by 3.

Solution:

If $P(n)$ respresent the above proposition then for $n=1$, $P(n)$ becomes,

$= 7^1 - 4^1$

$= 7 - 4$

$= 3$ Since 3 is exactly divisible by 3, the given statement is true for $n=1$.

Assuming that the result to be true for $n=k$ we get the hypothesis:

$7^k - 4^k$.

Now, we have to prove that the statement is true for $n=k+1$.

$7^{k+1} - 4^{k+1}$

$7.7^k - 4.4^k + \cancel{\ } - 3(4^k) + 3(4^k)$

$7\{7^k - 4^k\} + 3(4^k)$

By our hypothesis $(7^k - 4^k)$ is divisible by 3 and $3(4^k)$ is obviously ~~divis~~ divisible by 3. Therefore, the given expression is also divisible by 3 for $n=k+1$.

Hence, By mathematical Induction the given statement is true.

Q#3: Prove that:

i. $2^{n+1} > (2n+3)$, for all integral values of $n \geq 2$.

Proof:

If $P(n)$ represent the above proposition, then for $n=2$, $P(n)$ becomes:

$(2^{1+1}) > (2 \times 1 + 3)$

$2^2 > (2+3$

$4 > 5$

$$2^{2+1} > (2 \times 2 + 3)$$

$$2^3 > (4+3)$$

$$8 > 7$$

Since, $8 > 7$ which is true.

So $P(n)$ is true for $n=2$.

Assuming that statement is true for $n=K$ we get hypothesis:

$$2^{K+1} > 2K+3$$

Multiplying both sides of the above inequality by 2, we get,

$$2.2^{K+1} > 2(2K+3)$$

$$2^{K+1+1} > 4K+6$$

$$2^{K+1+1} > 4K+2+4$$

$$2^{K+1+1} > 2K+2K+2+4$$

$$2^{K+1+1} > 2K+2+3+2K+1$$

$$2^{K+1+1} > [2(K+1)+3] + 2K+1$$

Deducting $2K+1$ from right side, the inequality statement holds we have.

$$2^{K+1+1} > [2(K+1)+3]$$ $P(n)$ is true for $n=K+1$

Hence, by the principle of mathematical Induction $P(n)$ is true for all integral values of $n \geq 2$.

ii. $3^{n-1} > 2^n$, for all integral values of $n \geq 3$.

Proof:

If $P(n)$ represent the above proposition, then

For $n = 3$, $P(n)$ becomes.

$$3^{3-1} > 2^3$$

$$3^2 > 2^3$$

$9 > 8$ Since $P(n)$ is true for $n = 3$.

Assuming that statement is true for $n = k$, we get hypothesis.

$$3^{k-1} > 2^k$$

Multiplying both sides of the above inequality by 3.

$$3.3^{k-1} > 3.2^k$$

$$3^{k+1-1} > 2^k(3)$$

$$3^{k+1-1} > 2^k(2+1)$$

$$3^{k+1-1} > 2^k.2 + 2^k$$

$$3^{k+1-1} > 2^{k+1} + 2^k$$

Deducting 2^k from right sides, the inequality statement holds. we have.

$$3^{k+1-1} > 2^{k+1}$$

Therefore $P(n)$ is also true for $n = k+1$

Hence by the mathematical Induction $P(n)$ is true for all integral values of $n \geq 3$

$3^{n-1} > 2^n$

$3^{3-1} > 2^3$

$3^2 > 8$

$9 > 8$

2.2^k

2^{k+1}

iii. $n! > 3^{n-1}$, for all integral values of $n \geq 5$.

Proof:

If $P(n)$ represent the above proposition then

For $n=5$, $P(n)$ becomes.

$$5! > 3^{5-1}$$

$$5 \times 4 \times 3 \times 2 \times 1 > 3^4$$

$120 > 81$ $\therefore P(n)$ is true for $n=5$

Assuming that statement is true for $n=k$, we get hypothesis:

$$k! > 3^{k-1}$$

multiplying by both sides by $(k+1)$:

$$(k+1)\,k! > (k+1)\,3^{k-1}$$

$$(k+1)! > (k+3-2)\,3^{k-1}$$

$$(k+1)! > (3^{k-1})(3-2+k)$$

$$(k+1)! > 3.3^{k-1} + (3^{k-1})(-2+k)$$

$$(k+1)! > 3^{k+1-1} + 3^{k-1}(-2+k)$$

Deducting $3^{k-1}(-2+k)$ from right side. the inequality statement holds and we have.

$(k+1)! > 3^{k+1-1}$ $\therefore P(n)$ is true for $n=k+1$

Hence, By priciple of mathematical Induction $P(n)$ is true for all positive integral values of $n \geq 5$.